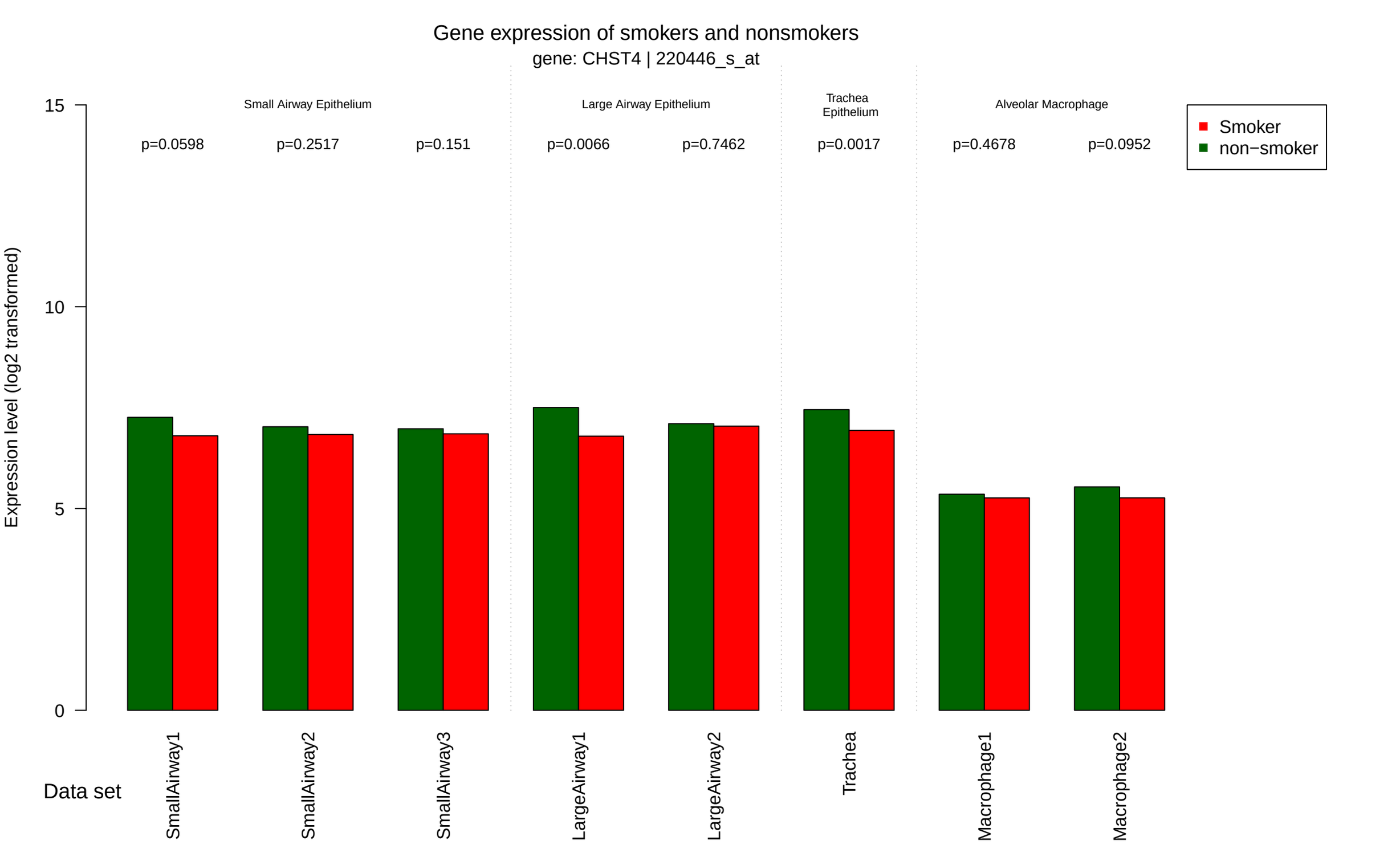
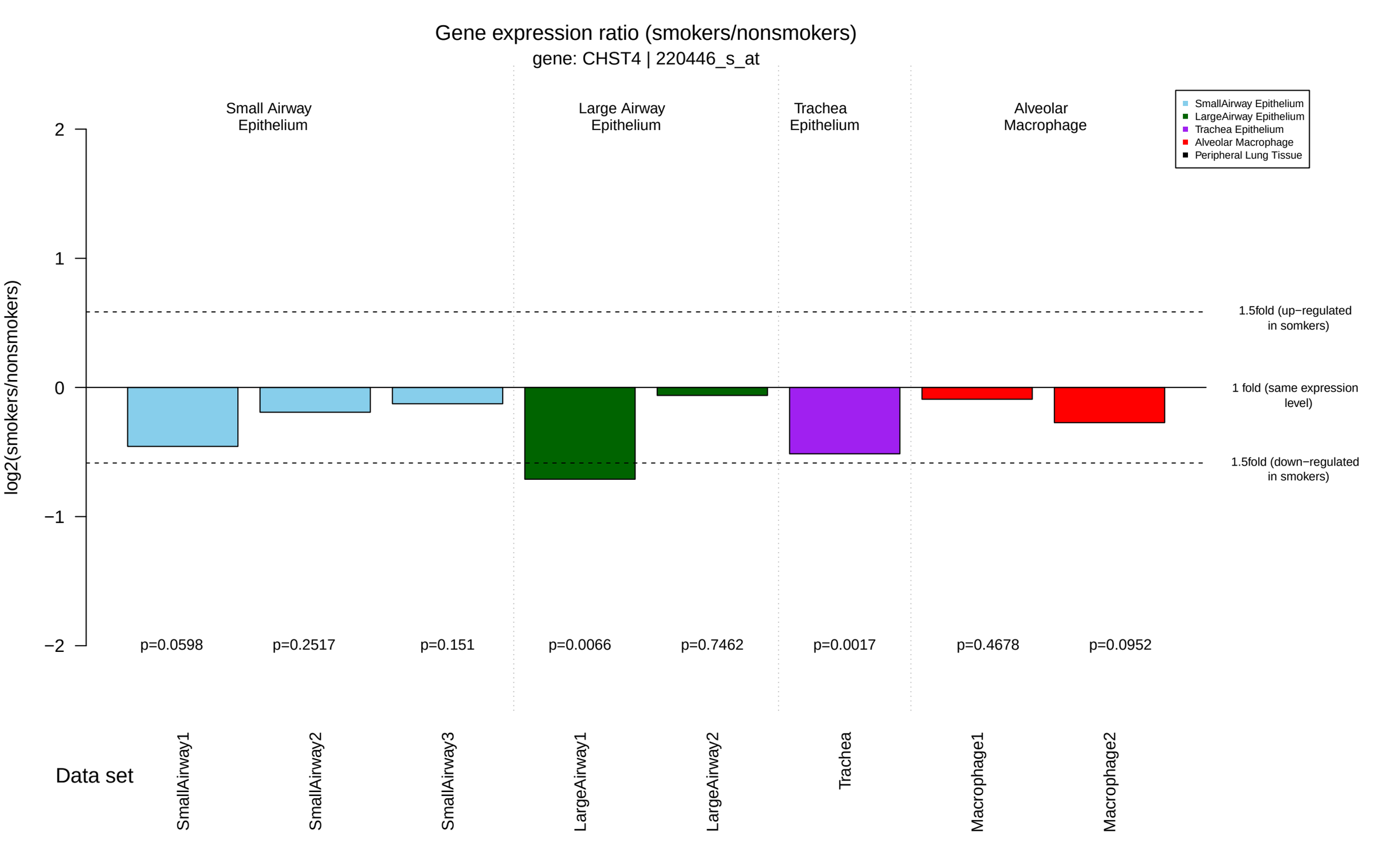
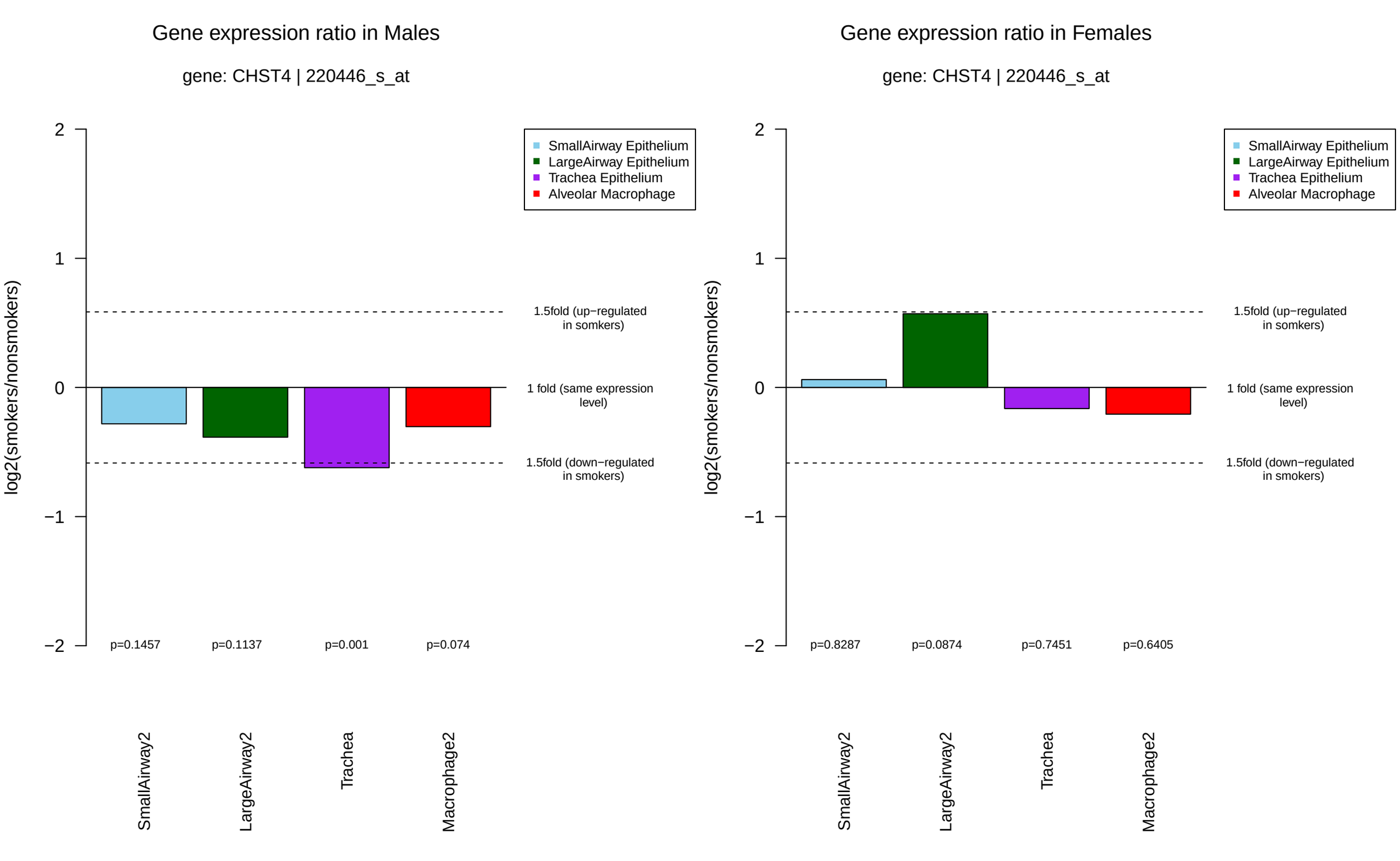
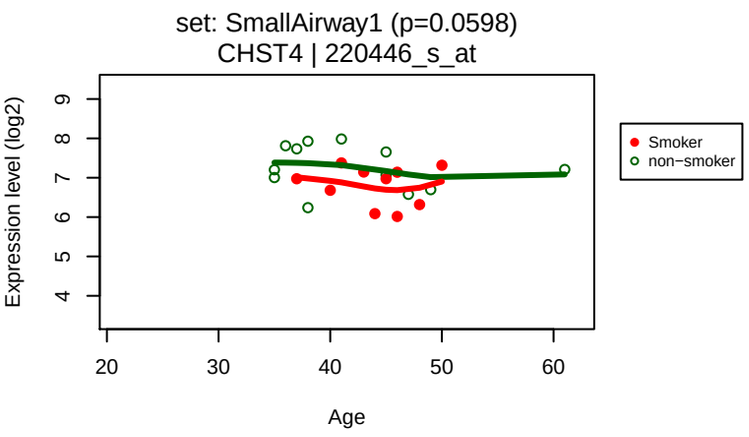
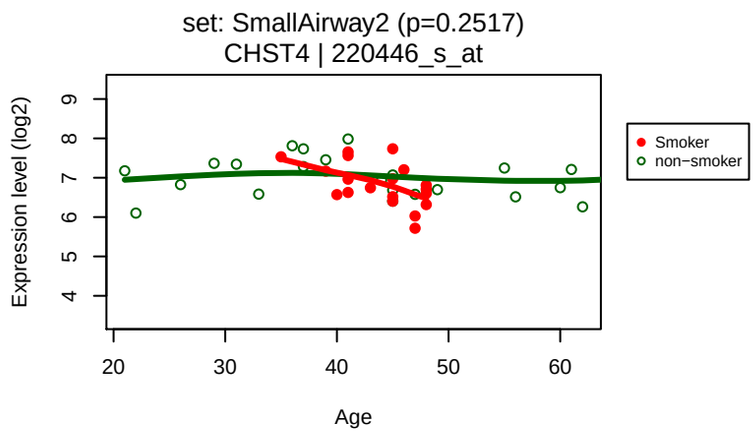
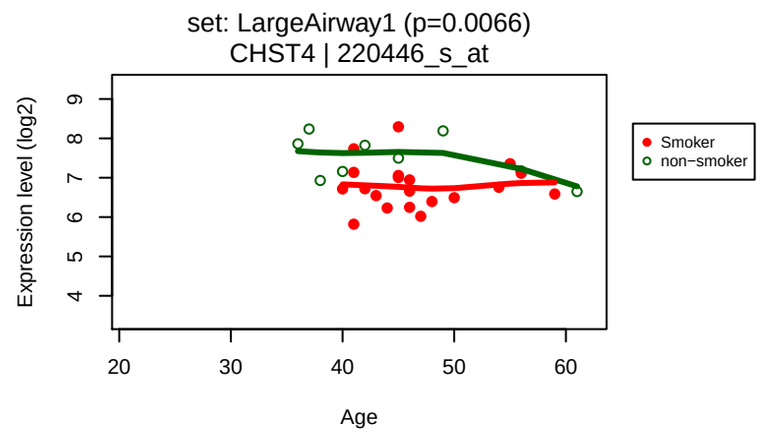

Gene expression of smokers and nonsmokers
gene: CHST4 | 220446_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0598
p=0.2517
p=0.151
p=0.0066
p=0.7462
p=0.0017
p=0.4678
p=0.0952
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

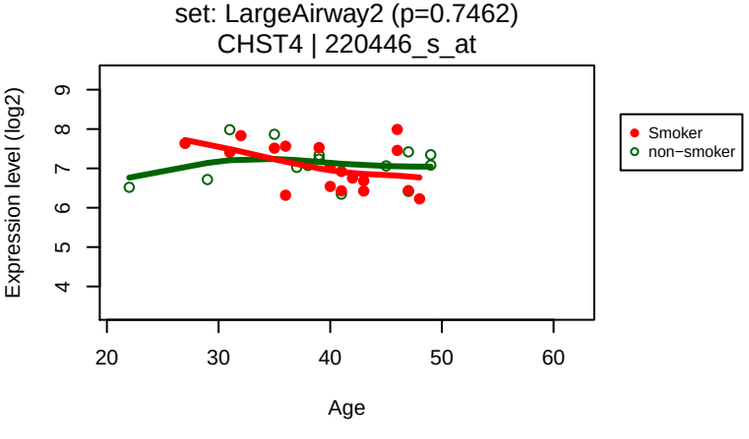

Gene expression ratio (smokers/nonsmokers)
gene: CHST4 | 220446_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0598
p=0.2517
p=0.151
p=0.0066
p=0.7462
p=0.0017
p=0.4678
p=0.0952
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

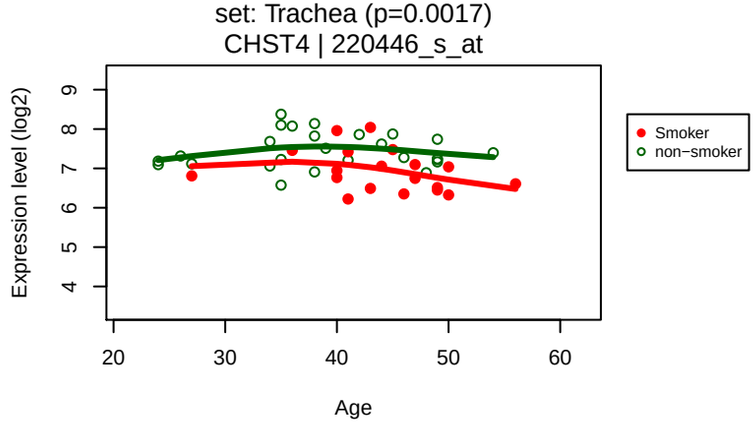

Gene expression ratio in Males
gene: CHST4 | 220446_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1457
p=0.1137
p=0.001
p=0.074
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHST4 | 220446_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8287
p=0.0874
p=0.7451
p=0.6405
SmallAirway2
LargeAirway2
Trachea
Macrophage2

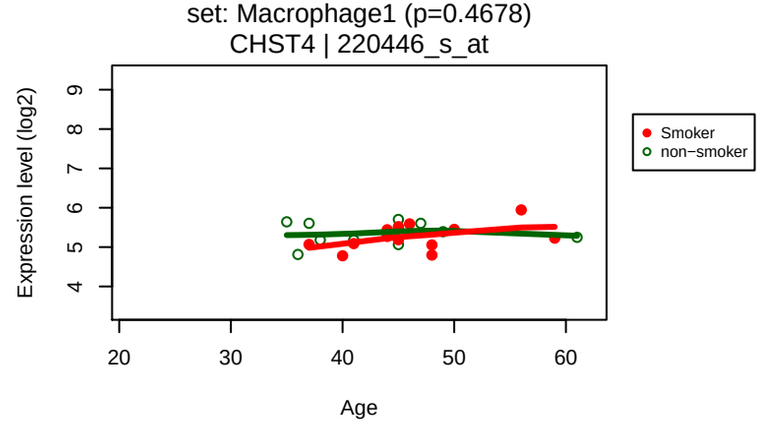

set: SmallAirway1 (p=0.0598)
CHST4 | 220446_s_at
Expression level (log2)
Age
Smoker
non−smoker

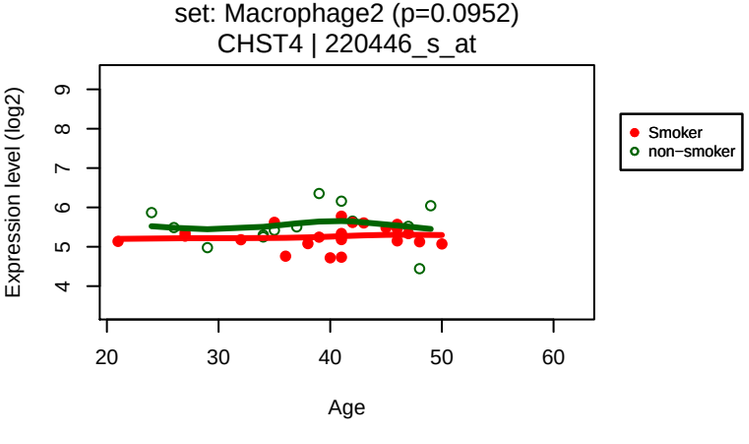

set: SmallAirway2 (p=0.2517)
CHST4 | 220446_s_at
Expression level (log2)
Age
Smoker
non−smoker

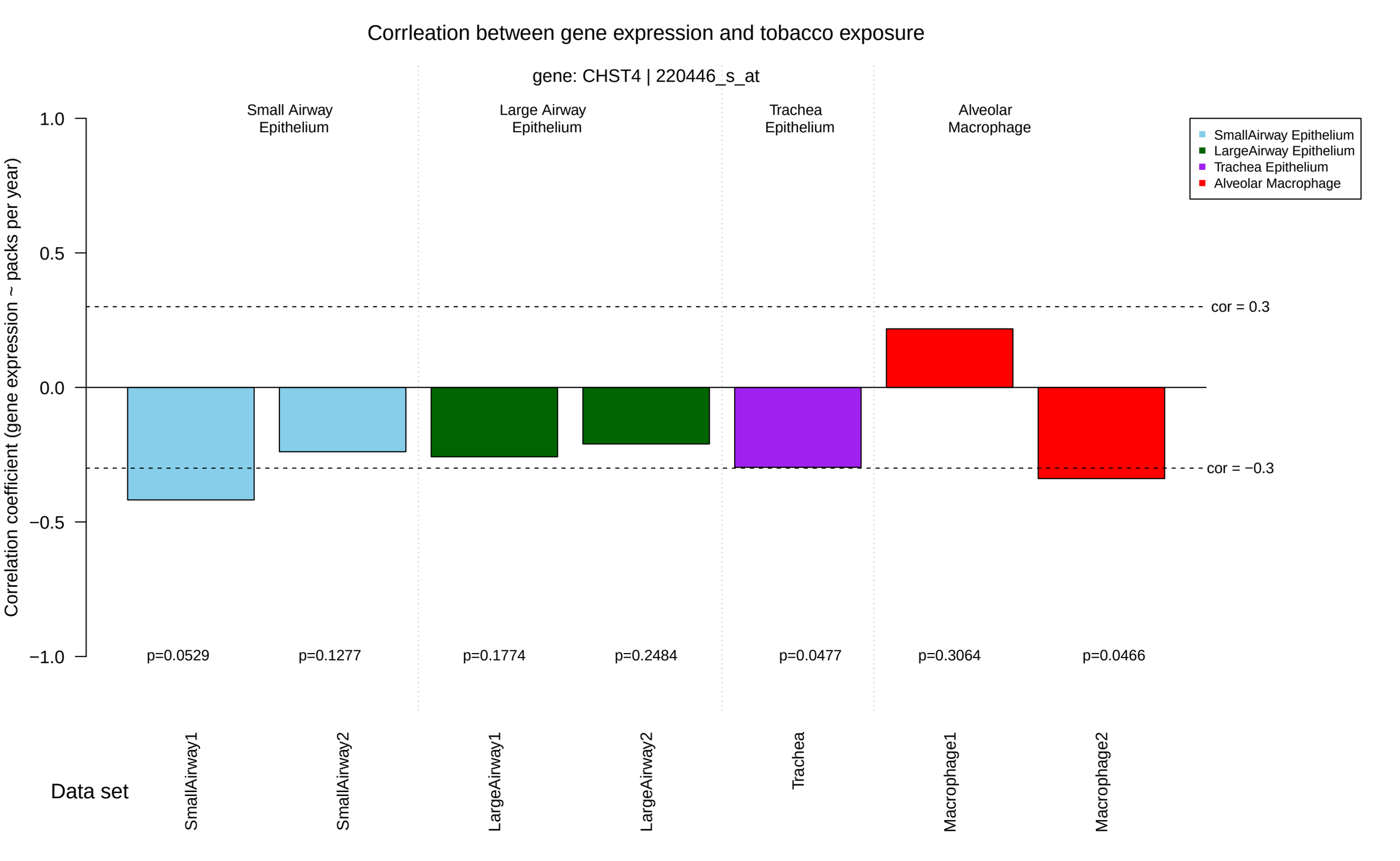

set: LargeAirway1 (p=0.0066)
CHST4 | 220446_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7462)
CHST4 | 220446_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0017)
CHST4 | 220446_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4678)
CHST4 | 220446_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0952)
CHST4 | 220446_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHST4 | 220446_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0529
p=0.1277
p=0.1774
p=0.2484
p=0.0477
p=0.3064
p=0.0466
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2